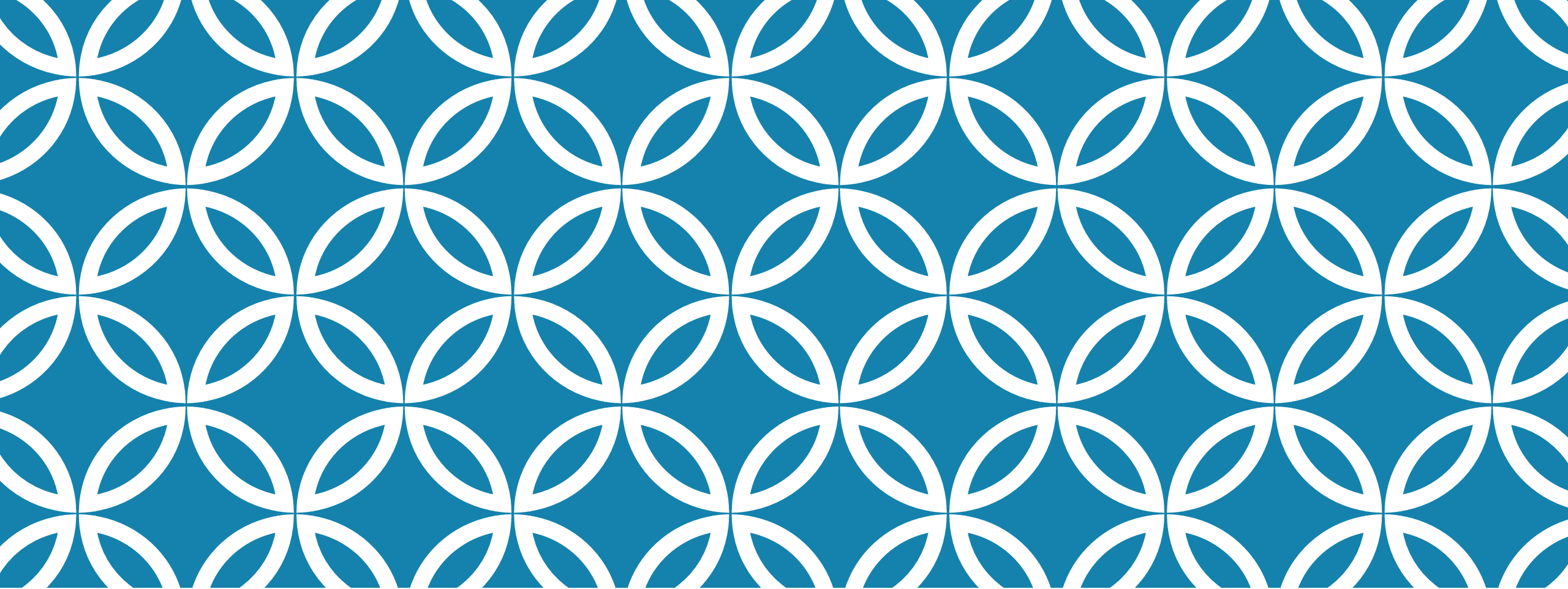




DATA ANALYSIS

GYM MEMBER EXERCISE

Project By: Nizam Jaafar
LinkedIn: [mdnizamjaafar](#)
Github: [mdnizamjaafar](#)

OVERVIEW

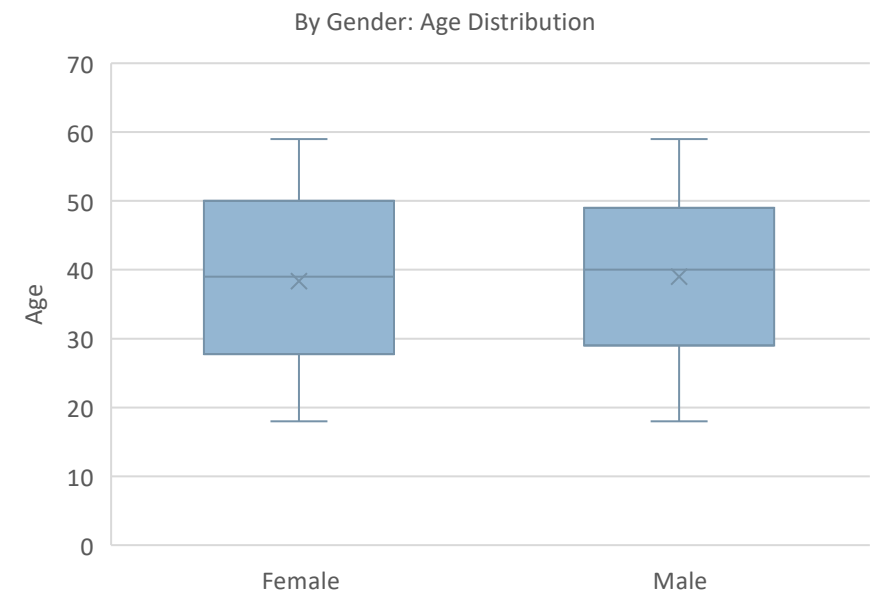
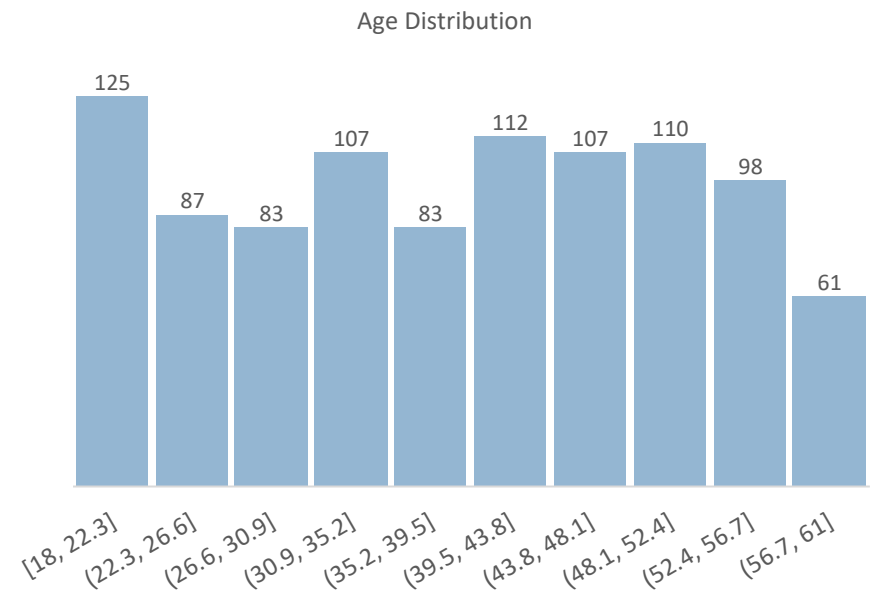
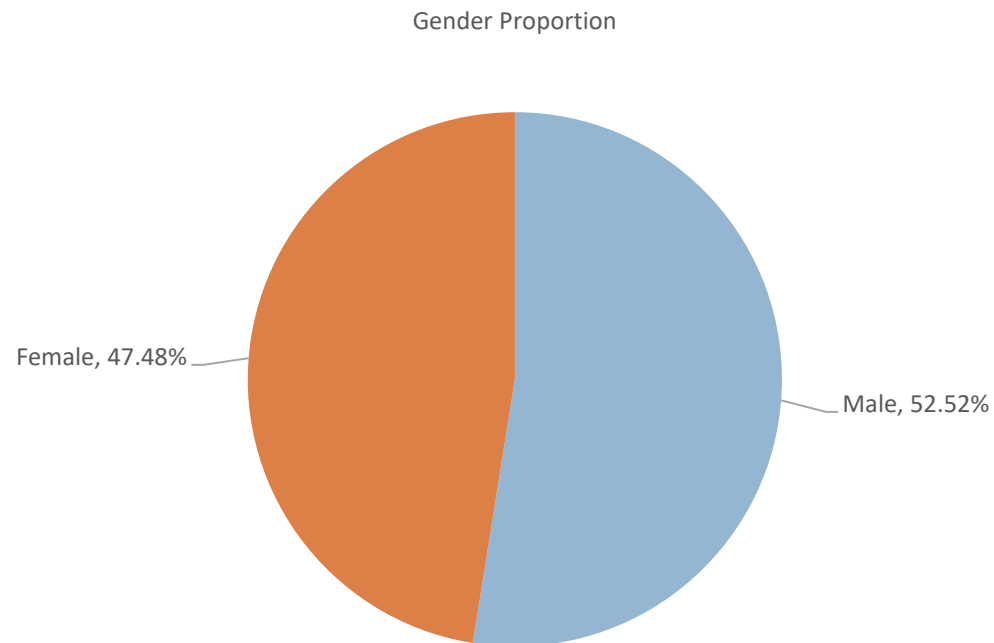
This is an analysis project on gym members' fitness, health and performance patterns and trends. The original dataset did not include crucial information on BMI and fat classification, which are essential for extensive research on gym members' overall health. Therefore, an extra initiative was taken to pull out the information using trusted sources and integrate it with the original dataset. Now, we can delve deeper into their health and have also classified their heart rate usage during exercise based on their age and etc.

Sources Used:

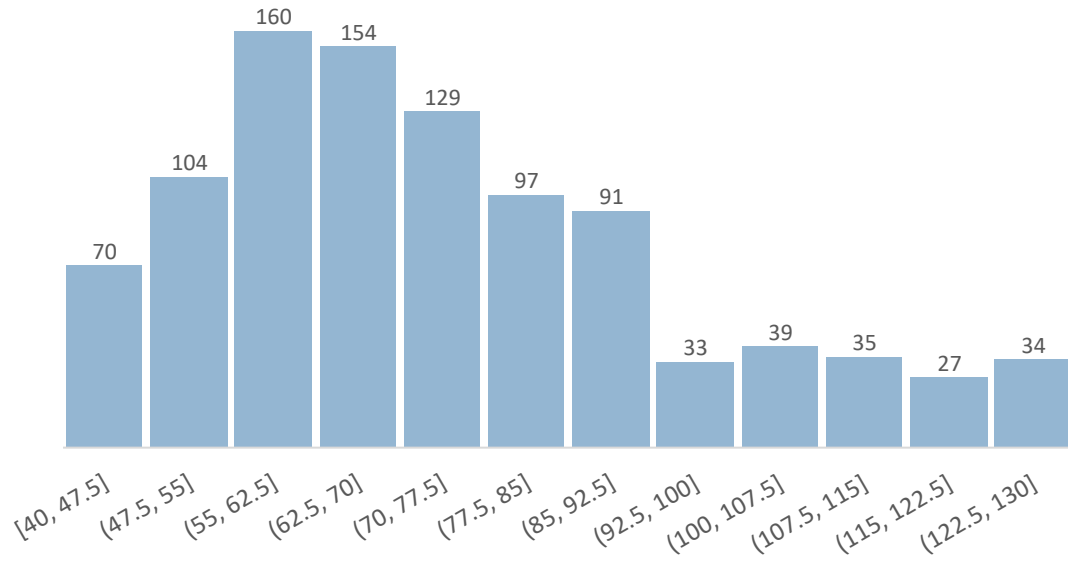
1. Dataset: [Kaggle](#)
2. BMI Classification: [National Center for Biotechnology Information](#)
3. Fat Classification: [Omron Healthcare](#)
4. Exercise Heart Rate Classification: [American Heart Association](#)
5. Resting Heart Rate Classification: [MedicineNet](#)

Goals:

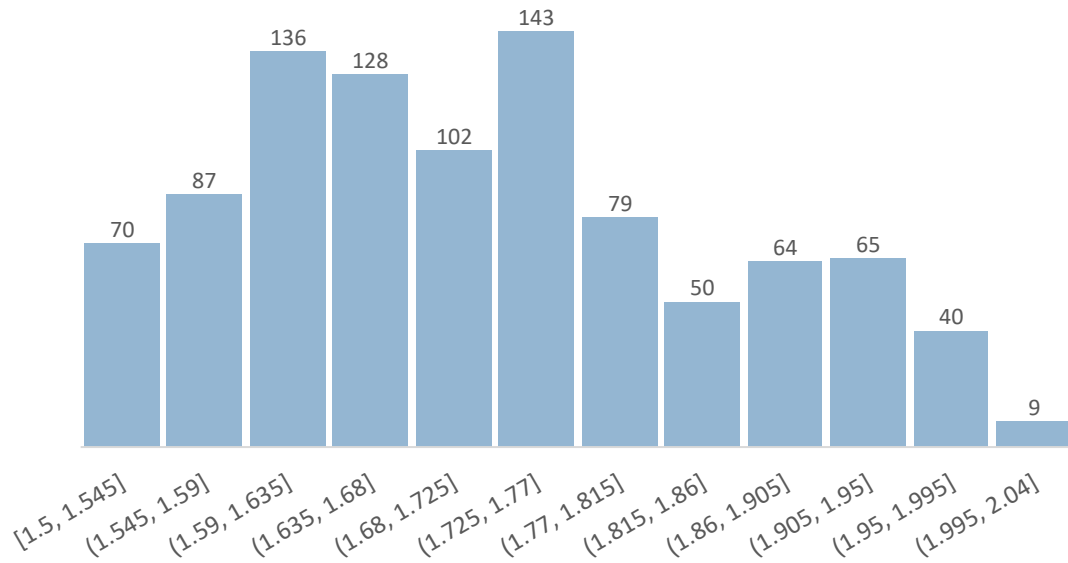
1. Provide insights into gym members' fitness, health and performance trends.
2. Analyze and report on correlation of each metrics.
3. Identify members who require attention for their overall health.
4. Offer insights into different workout type routines.
5. Provide proposal for fitness, health and performance improvement.



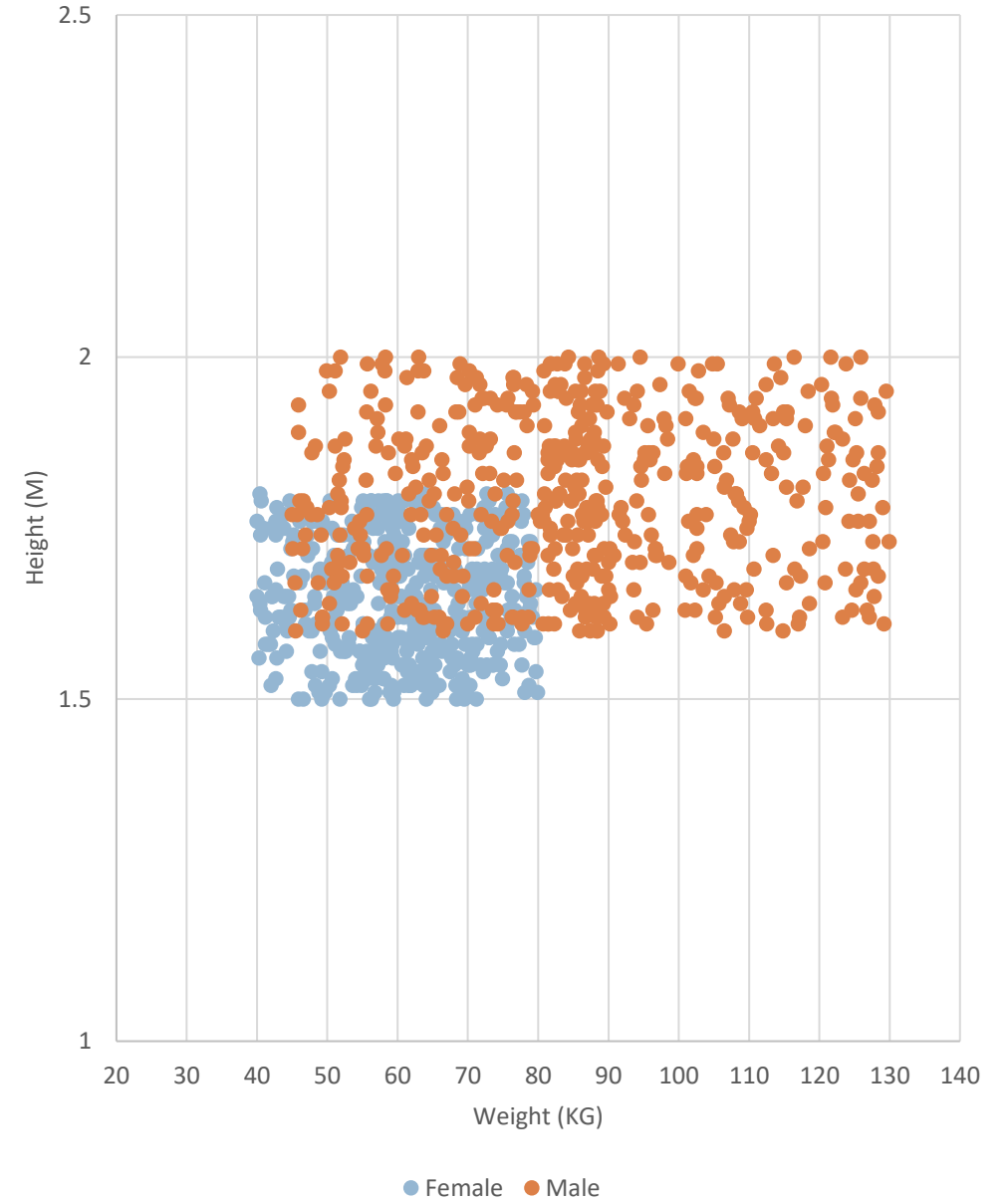
Weight Distribution

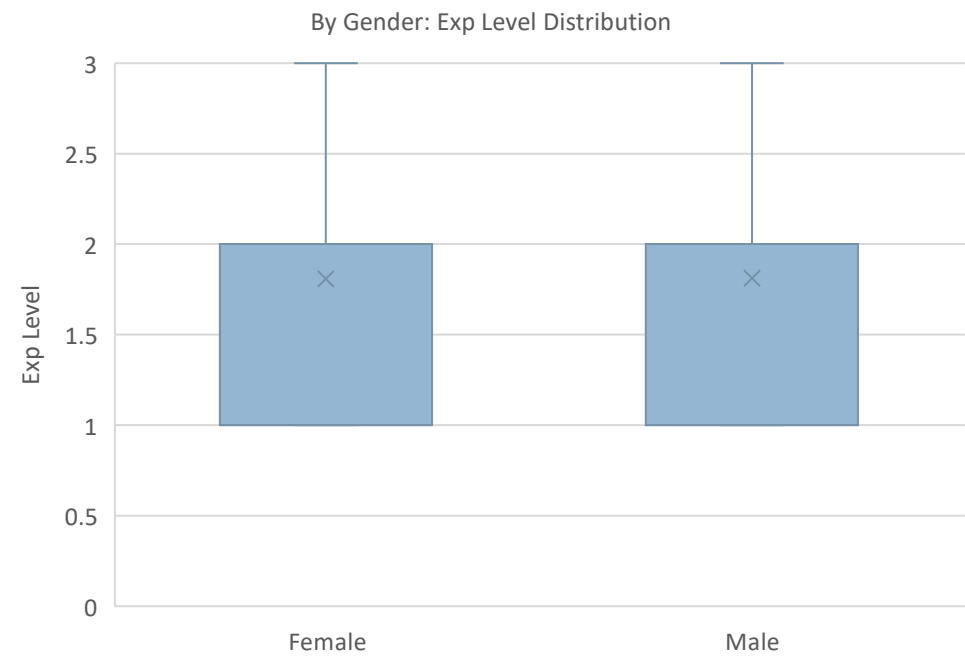
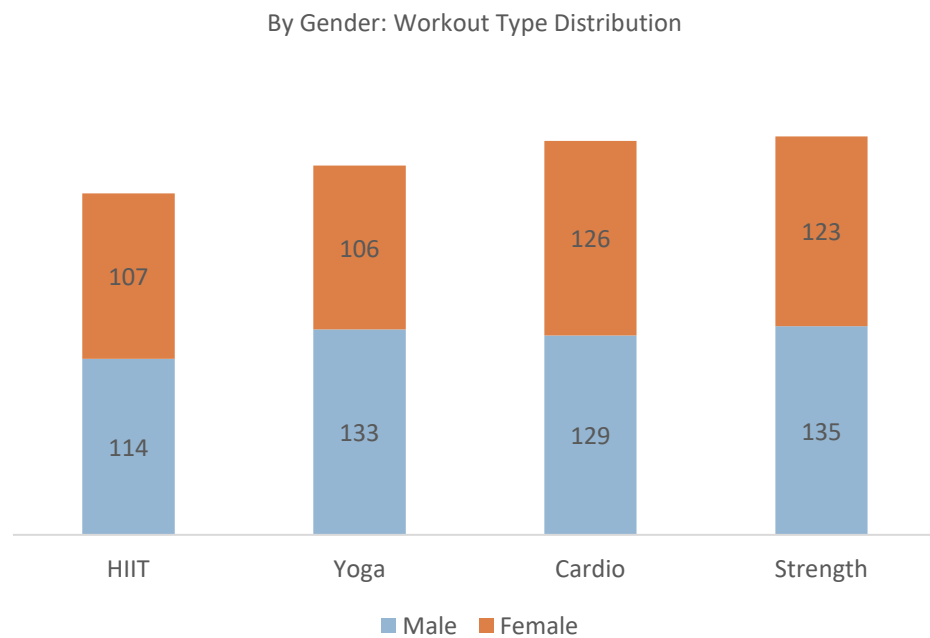
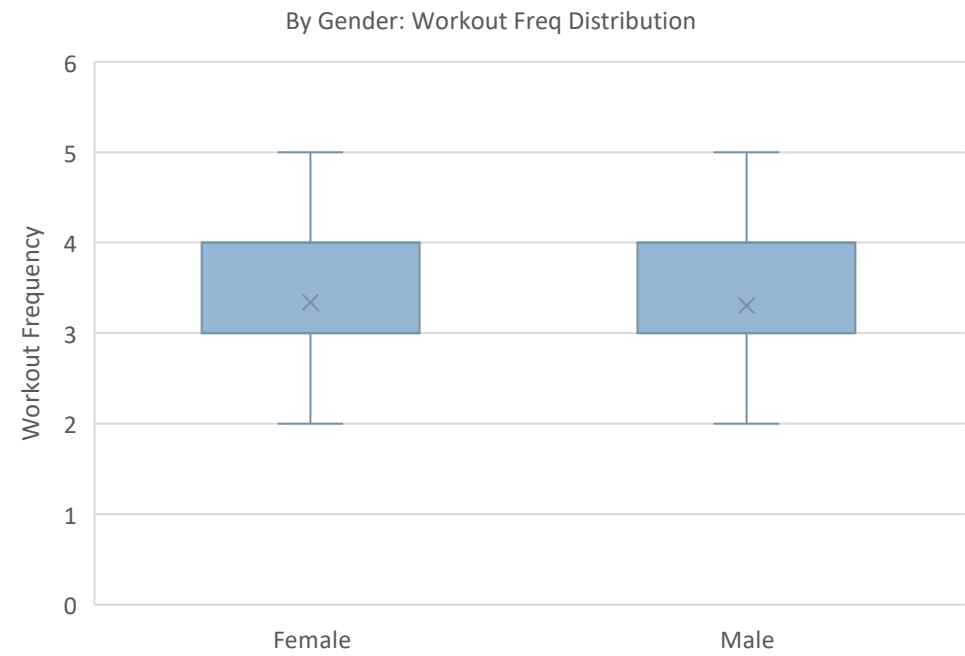
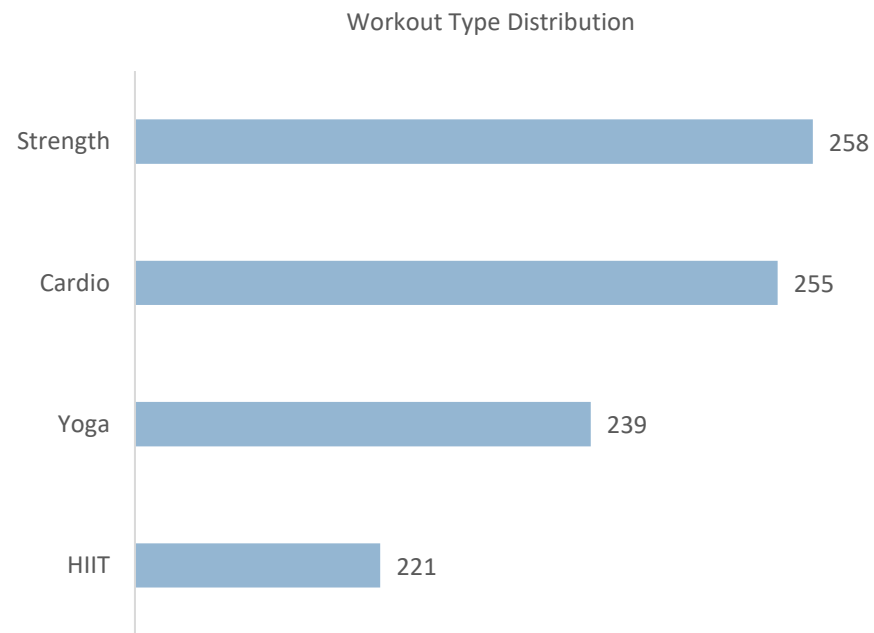


Height Distribution



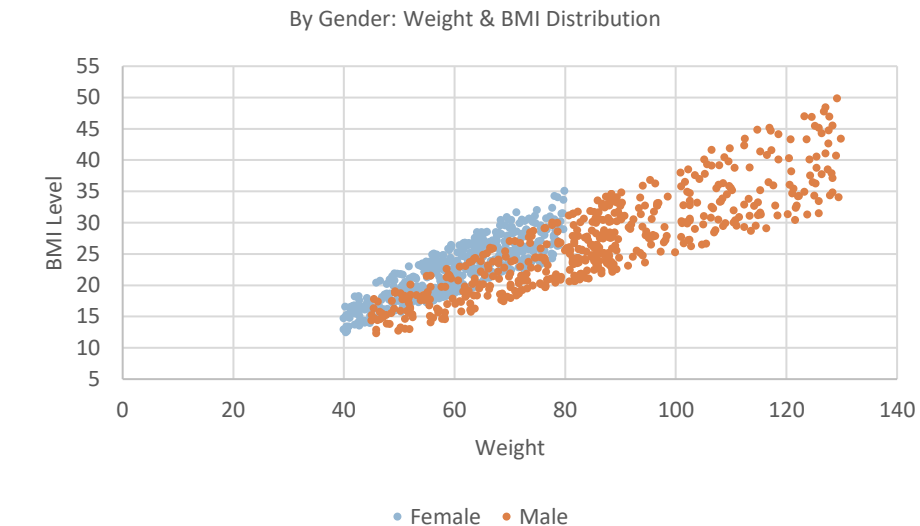
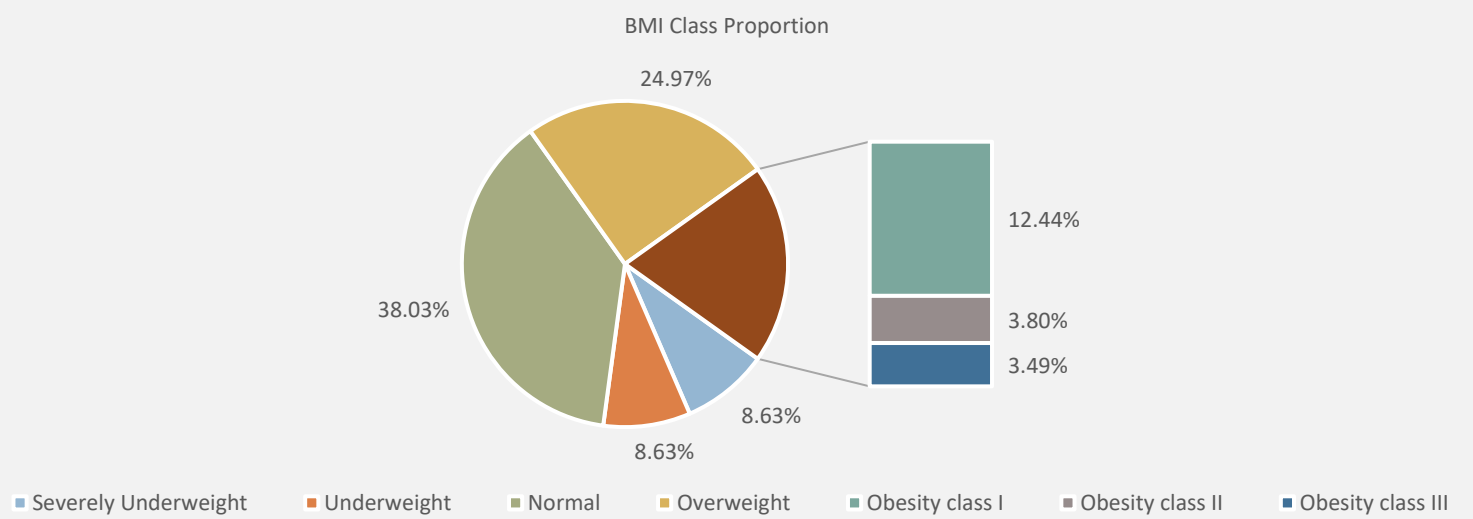
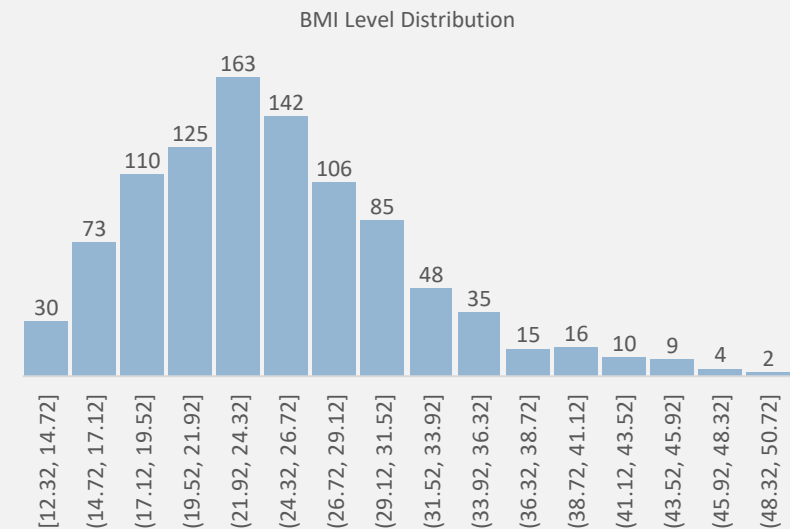
By Gender: Weight & Height Distribution





CORRELATION MATRICES

Variable	Age	Weight (KG)	Height (M)	BMI	Fat Percentage	Experience Level	Workout Frequency (Day / Week)	Session Duration (Hours)	Max BPM	Avg BPM	Resting BPM	Calories Burned
Age	1											
Weight (KG)	-0.036339635	1										
Height (M)	-0.027837495	0.365321203	1									
BMI	-0.01369137	0.85315769	-0.15946875	1								
Fat Percentage	0.002370051	-0.22551164	-0.235520936	-0.11925776	1							
Experience Level	-0.018675927	0.003378528	-0.010266611	0.016031073	-0.654362613	1						
Workout Frequency (Day / Week)	0.008055163	-0.011769328	-0.011269883	0.001644974	-0.537059548	0.837078709	1					
Session Duration (Hours)	-0.019911904	-0.013665561	-0.010205897	-0.006492647	-0.581519771	0.764768119	0.644140366	1				
Max BPM	-0.017072597	0.05706113	-0.017659884	0.067105231	-0.009055731	0.000544834	-0.029099066	0.010050981	1			
Avg BPM	0.035969143	0.009717478	-0.014776288	0.0216055	-0.007301655	-0.000888157	-0.010680798	0.016014438	-0.039751443	1		
Resting BPM	0.004353714	-0.032138091	-0.005089864	-0.032542632	-0.016834389	0.001757585	-0.007966891	-0.016648808	0.036647481	0.059635502	1	
Calories Burned	-0.15467876	0.095443473	0.086348051	0.059760826	-0.597615248	0.694129448	0.576150125	0.908140376	0.002090016	0.339658667	0.016517951	1
Water Intake (Liters)	0.041528359	0.39427571	0.393532902	0.213696572	-0.588682834	0.304103549	0.238562571	0.283410977	0.031620643	-0.002910637	0.007725998	0.356930683



Regression Statistics	
Multiple R	0.85315769
R Square	0.727878044
Adjusted R Square	0.727597795
Standard Error	3.476457142
Observations	973

Independent Variable	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	5.1220	0.4040	12.6782	0.0000	4.3292	5.9148	4.3292	5.9148
Weight (KG)	0.2680	0.0053	50.9633	0.0000	0.2576	0.2783	0.2576	0.2783

BMI Level Distribution

Skewed to the Right: Distribution mainly occurs around 12.32 to 36.32.

Proportion Analysis:

- Severely Underweight: 8.63%
- Underweight: 8.63%
- Normal: 38.03%
- Overweight: 24.97%
- Obesity Class: 19.73%

(Source: <https://www.ncbi.nlm.nih.gov/books/NBK541070/>)

Insight

1. Focus on Obesity & Underweight Class:

- Significant Proportion:** 19.73% falls under obesity class while underweight class around 17.26%.
- It's essential to understand that a higher or lower BMI doesn't necessarily equate to poor health.
- Additional Metrics:** Important to consider fat percentage and performance metrics for a comprehensive health assessment.

Regression Analysis

Statistically Significant: Strong relationship between BMI and Weight with a p-value under 0.5.

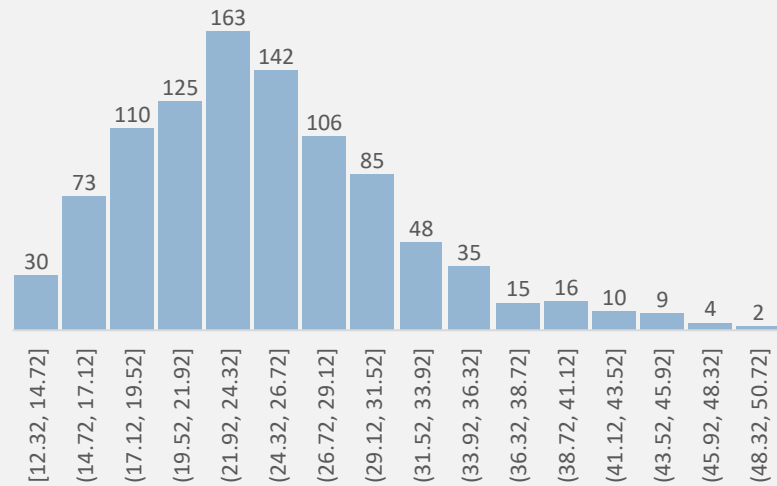
Variance Explained:

- R Square: 72.79% variance in BMI explained by Weight.

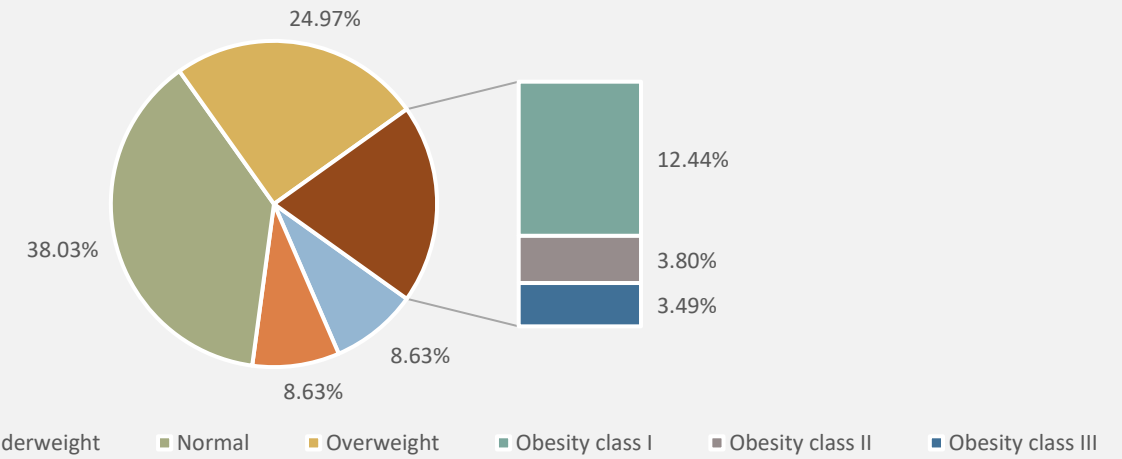
Coefficient:

- For every kilogram increase in Weight, BMI increases by approximately 0.268 units.

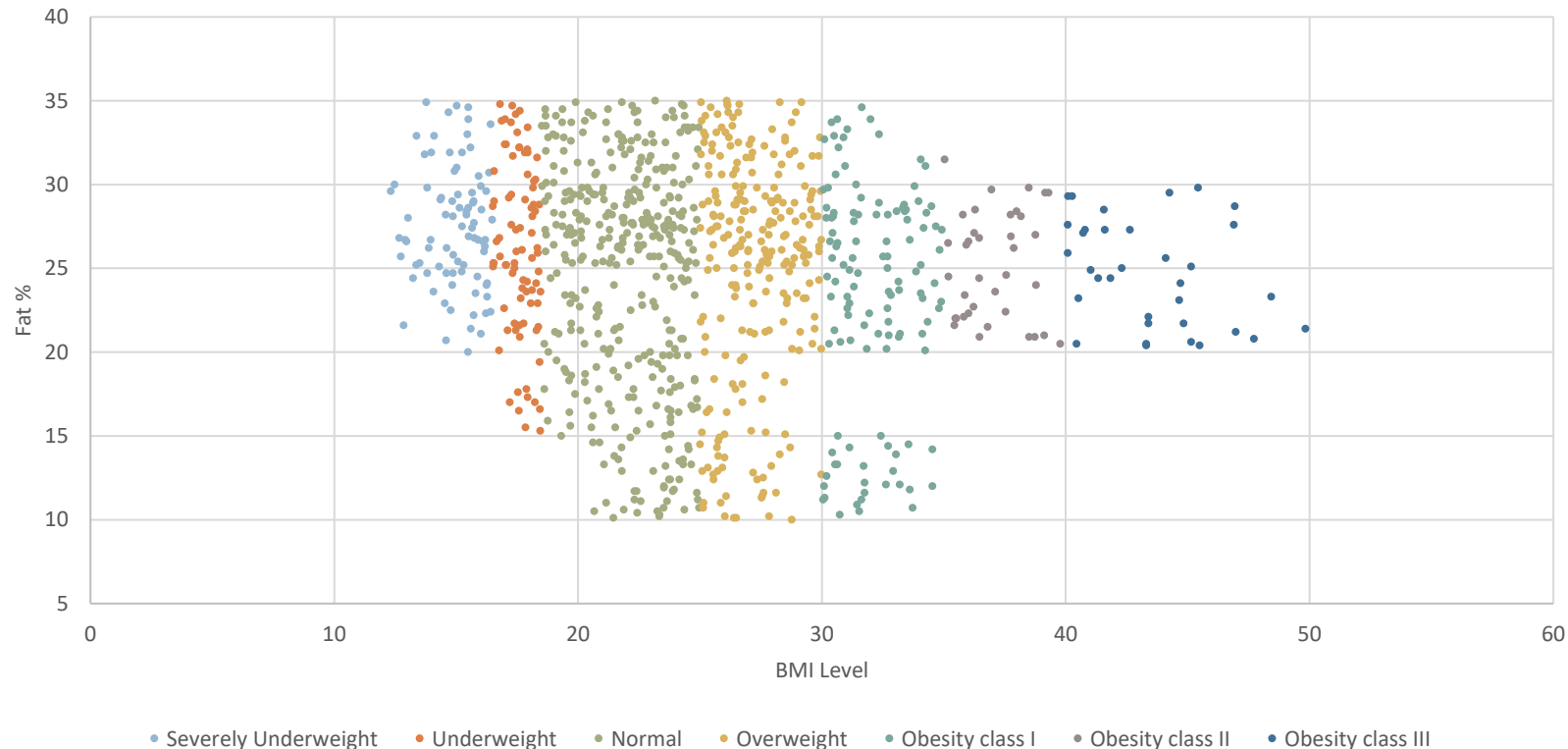
BMI Level Distribution



BMI Class Proportion



By BMI Class: BMI & Fat % Distribution



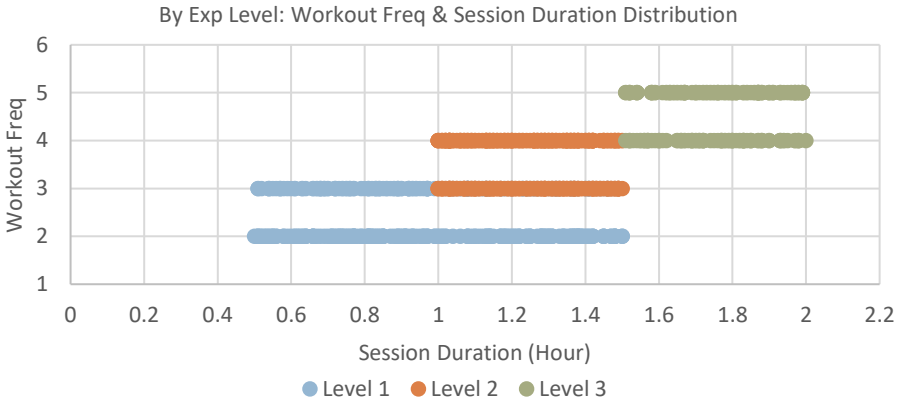
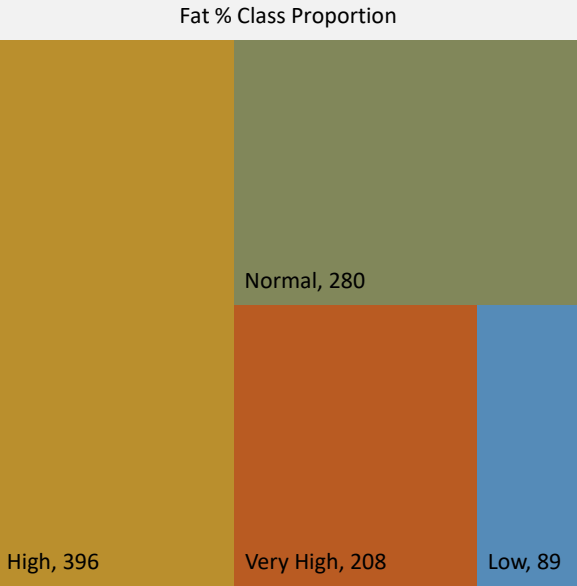
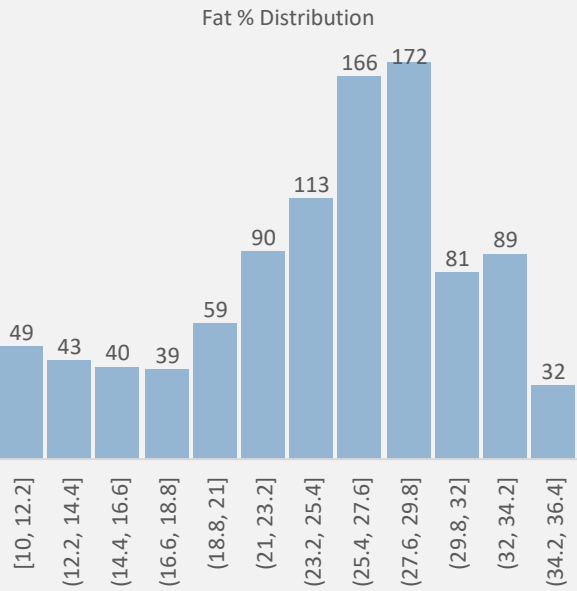
Obesity and Underweight Class

Insight:

- Analysis shows that for someone that categorize under class of Obesity or Underweight can still have better fat percentage than other BMI class.
- Many members in Obesity Class 1 and Overweight class have range of fat percentage around 10% to 15%. This reject the perception of BMI level indicate someone health status.
- Weight has a significant influence on BMI but weight can be impacted by bones and muscle density. (Source: <https://healthfully.com/bmi-index-according-to-bone-structure-6466650.html?form=MG0AV3>).
- Additionally, we can see for class of Normal, we found that more than 50% of gym members have fat percentage than 20%. The same goes to Underweight and Overweight Class

Proposal:

- To tailor and market a specialized fitness program for those with fat percentage than 20% for male members and above 30% for female members. (Source: <https://www.omronhealthcare-ap.com/my/faqs/weight-management>).
- Periodic health assessment to monitor their fat percentage by quarterly or monthly to have a further analysis on their improvement based on data collected.



Independent Variables	Coefficients	P-value
Intercept	36.81159415	4.7E-278
Experience Level	-4.54238299	8.37E-24
Workout Frequency (Day / Week)	0.262727999	0.381473
Session Duration (Hours)	-3.570782462	1.72E-07

Insight

1. Experience Level

Coefficient: -4.542, indicating a strong negative relationship with Fat Percentage. Each increase in experience level decreases Fat Percentage by approximately 4.54 units.

P-value: Extremely significant (8.36796E-24).

2. Workout Frequency (Days/Week)

Coefficient: 0.263, showing a very small positive relationship with Fat Percentage, but it's not statistically significant (P-value: 0.381).

3. Session Duration (Hours)

Coefficient: -3.571, showing a significant negative relationship with Fat Percentage. Each additional hour of session duration decreases Fat Percentage by approximately 3.57 units.

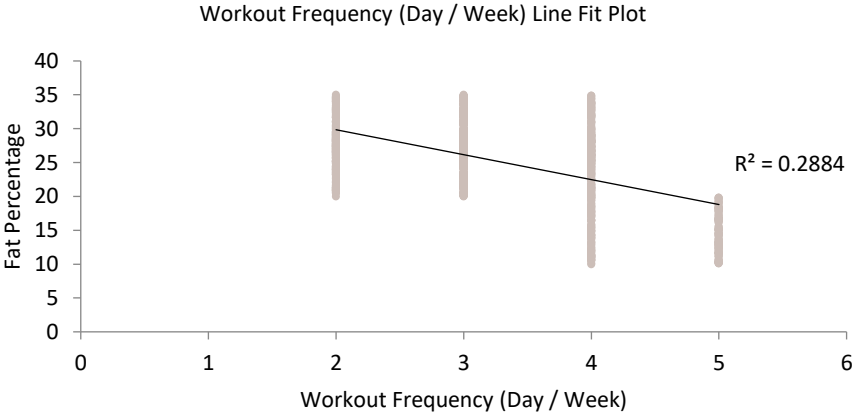
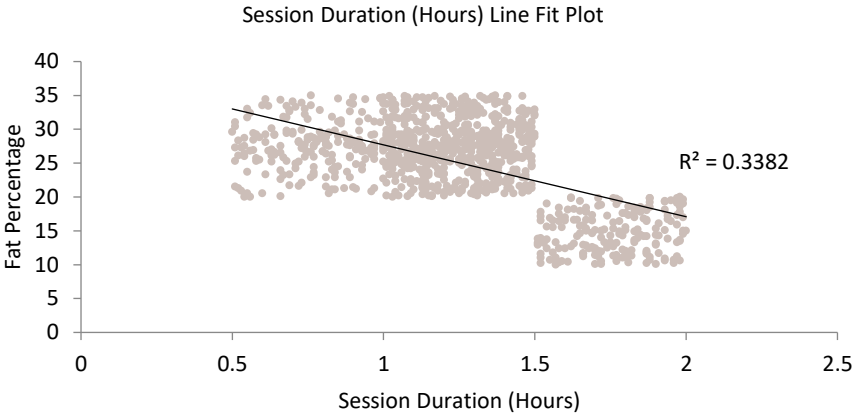
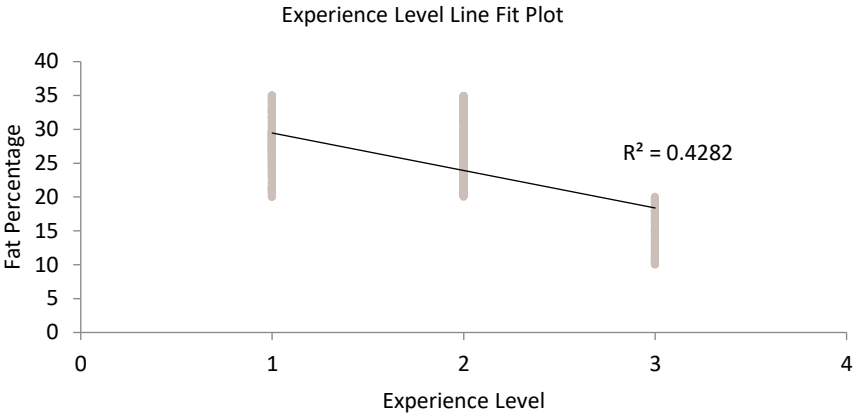
P-value: Highly significant (1.72249E-07).

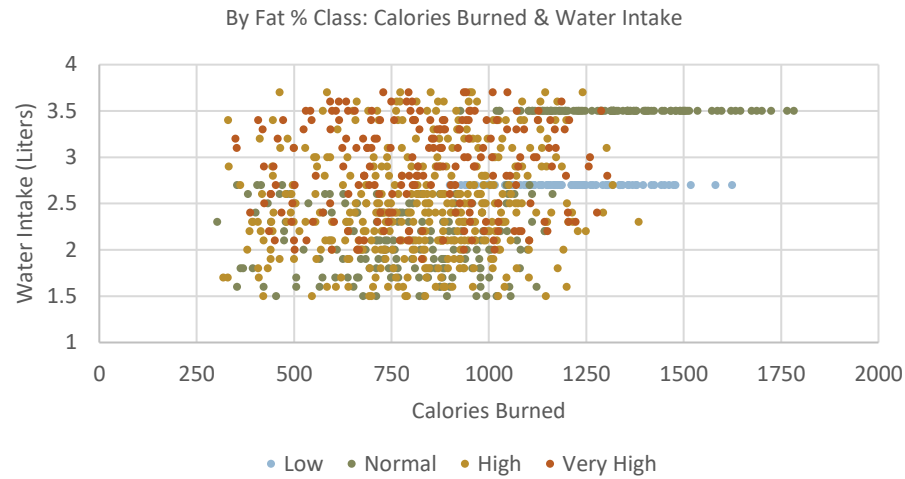
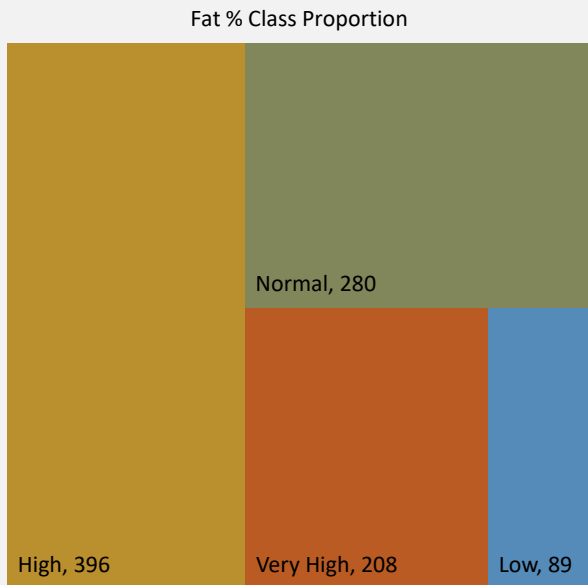
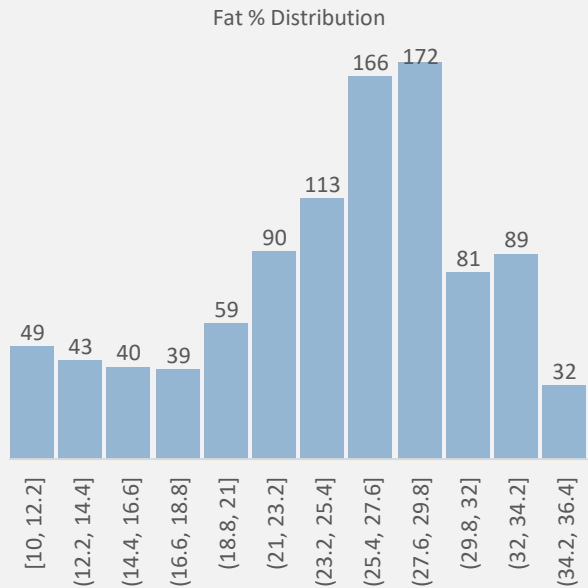
Conclusion

Experience Level and Session Duration significantly impact Fat Percentage, with higher levels and longer sessions leading to lower fat percentages. Workout Frequency, however, does not show a significant relationship in this multiple regression context.

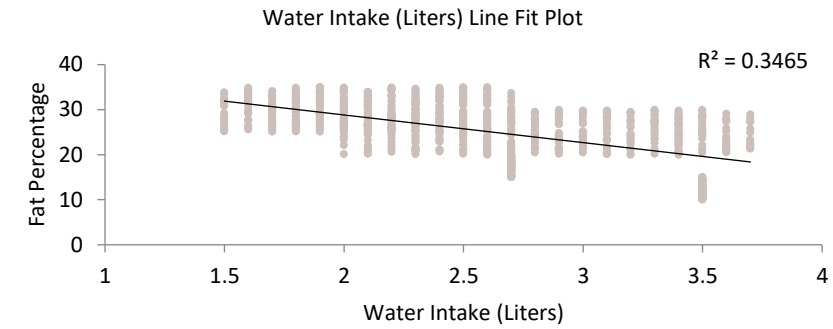
Proposal

- To strategize a program to enhance gym members experience and session duration with consultation from expert trainer.
- Market and campaign to emphasize on importance of exercise in improving fat percentage which directly asking for gym members and outsiders to exercise often; potentially increasing gym membership and revenue.





Independent Variables	Coefficients	P-value
Intercept	45.99206	0
Calories Burned	-0.0102	2.11E-66
Water Intake (Liters)	-4.48651	5.99E-63



Calories Burned & Water Intake Insight

- Members with low fat percentage tend to drink more water though the calories burned below 1000.
- Normal class fat percentage, will have higher water intake when they burned their calories above 1000.

Insight

1. Water Intake and Fat Percentage

Significance: Water intake has a significant relationship with fat percentage, with a p-value indicating strong statistical significance. For each liter of water consumed, fat percentage decreases by approximately 4.49 units.

Variance Explained: Only 34.65% of the variation in fat percentage can be explained by water intake, suggesting other factors are also at play.

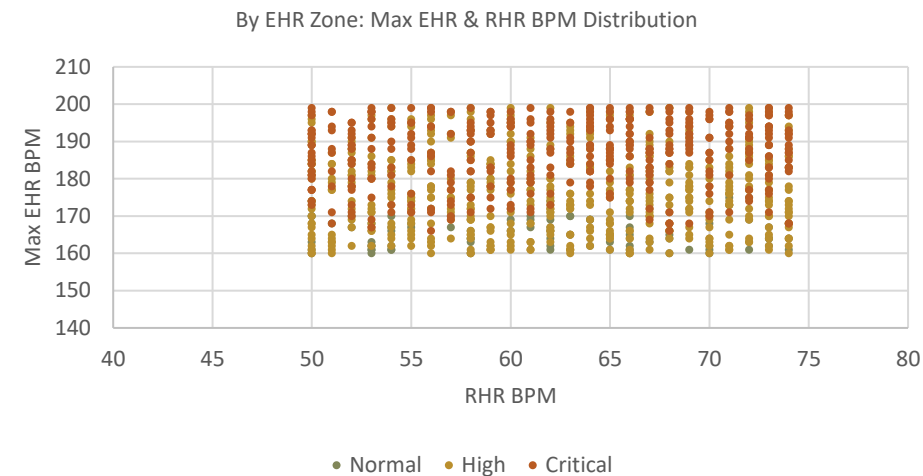
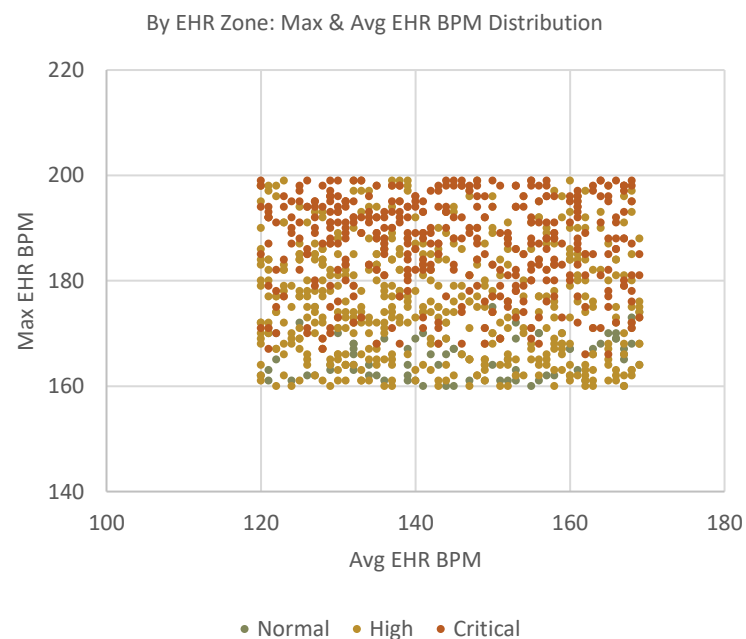
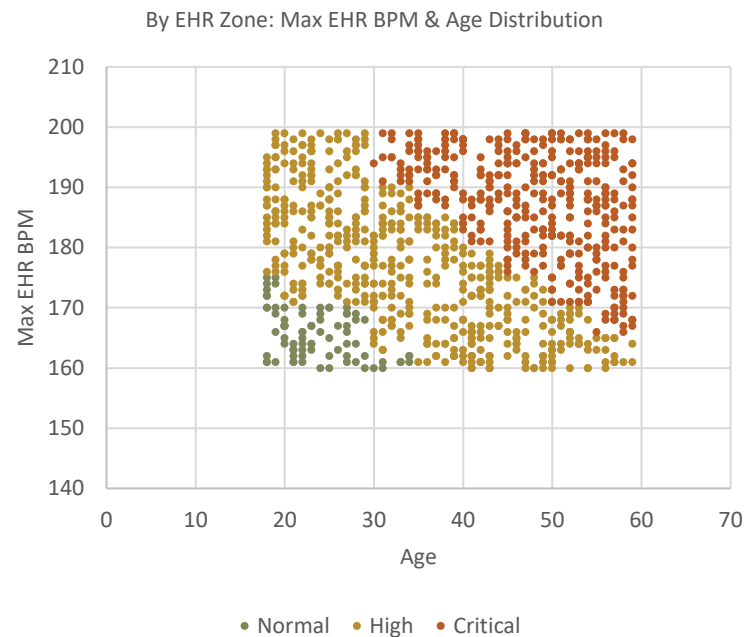
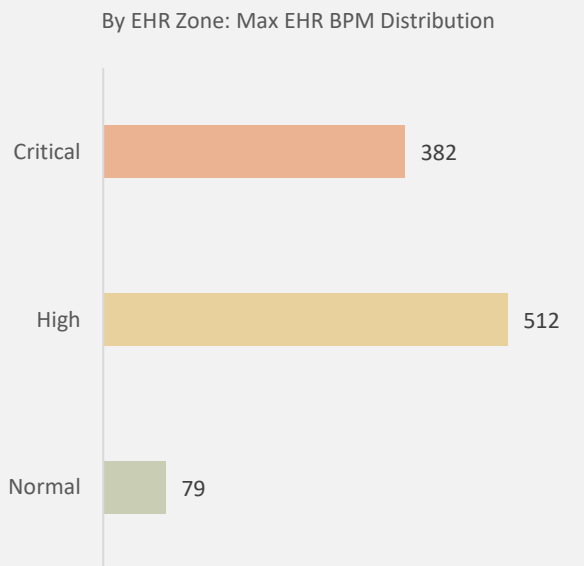
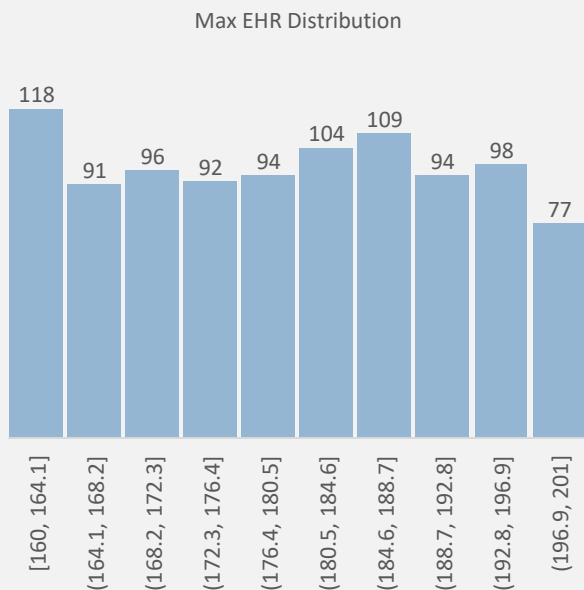
2. Calories Burned and Fat Percentage

Significance: Calories burned also show a significant relationship with fat percentage, with a p-value indicating strong statistical significance. Each calorie burned reduces fat percentage by approximately 0.0102 units.

Variance Explained: Only 35.71% of the variation in fat percentage can be explained by calories burned, indicating the influence of other variables on fat loss.

Proposal

- Encourage and guide members to increase their calories burning with appropriate exercise or workout routine.
- Plan a program that incorporate hydration strategies and effective workout with high calories burning to optimize fat loss.



	Max BPM
Age	-0.017072597
BMI	0.067105231
Fat Percentage	-0.009055731
Experience Level	0.000544834
Session Duration (Hours)	0.010050981

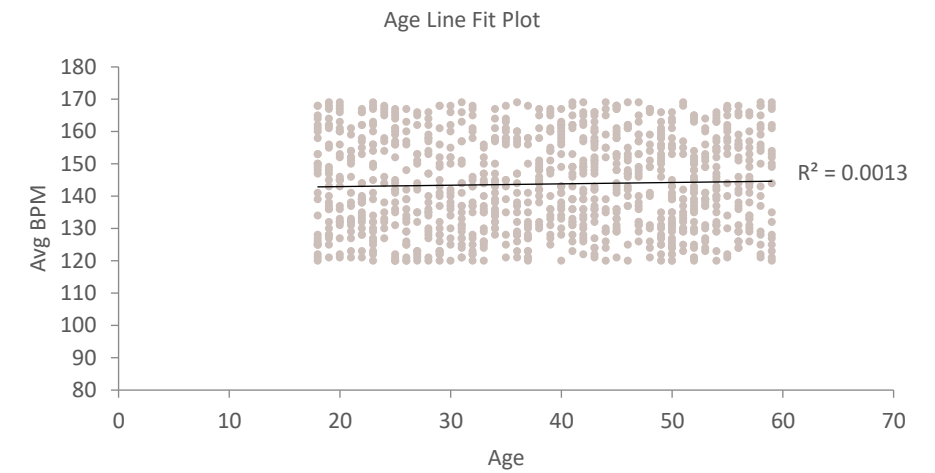
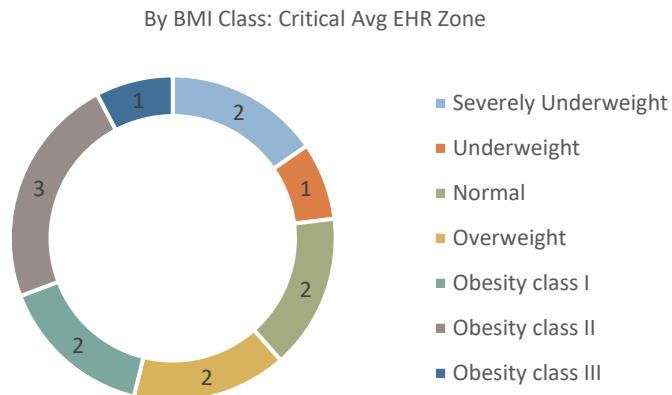
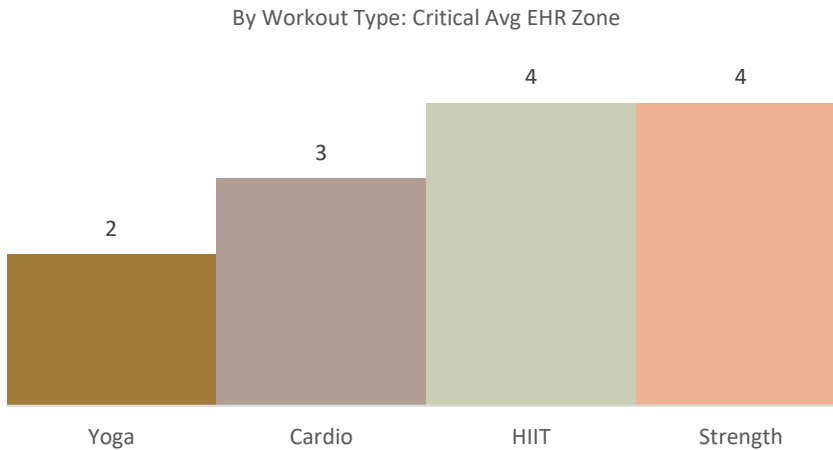
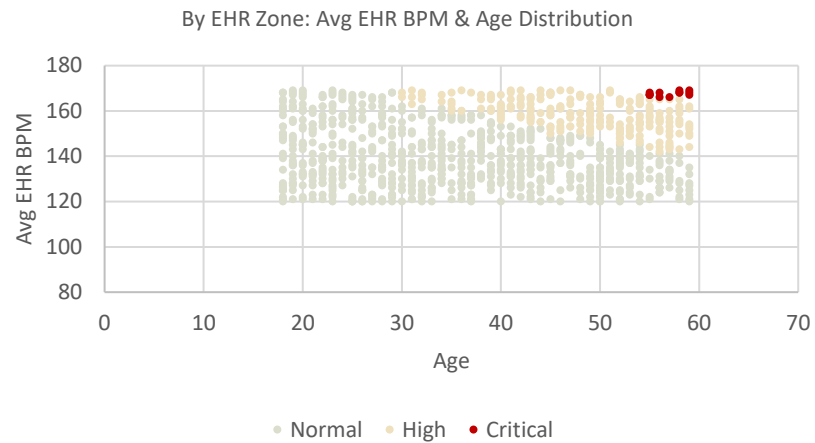
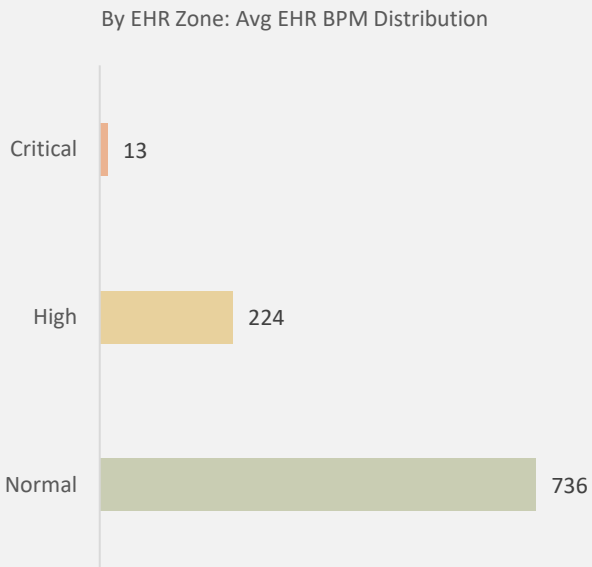
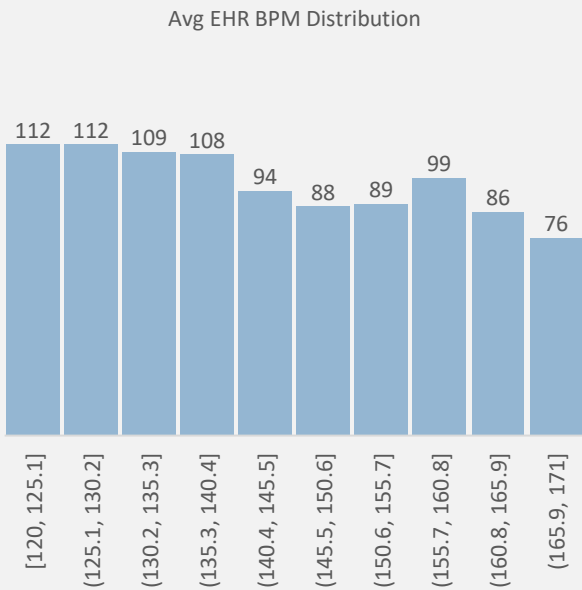
	Max BPM
Avg BPM	-0.039751443
Resting BPM	0.036647481
Calories Burned	0.002090016
Water Intake (Liters)	0.031620643

Insight

- The distribution of Max EHR BPM is evenly distributed with no skew.
- 91.88% gym members reach high and critical EHR zones during their workout.
- From the correlation matrices we can see that Max BPM almost have no correlation with all the demographic, behavioral, performance and fitness metrics.
- Scientifically, max exercise heart rate (EHR) BPM factored also by medication, stress, food and also environmental.
- To study more on gym members max EHR BPM, we should take extensive data collection on above factors.
- But, if we delve into age factor and referring to research from American Heart Association, EHR zone is impacted by age of person. The older they are, the lower their max zone level. (Source: <https://www.heart.org/en/healthy-living/fitness/fitness-basics/target-heart-rates>).

Proposal

- Conduct health assessment periodically for gym members to track their overall performance and fitness metric.
- Educate gym member on importance of monitoring heart level; how to ensure they are not jeopardizing their health during workout.



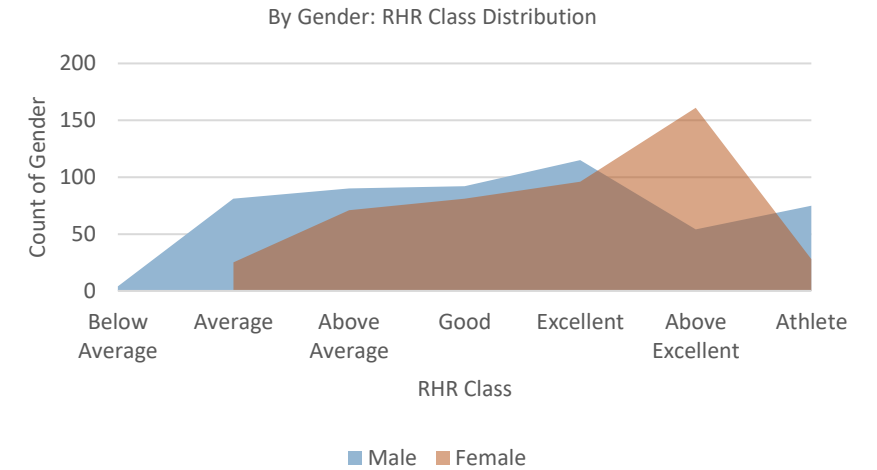
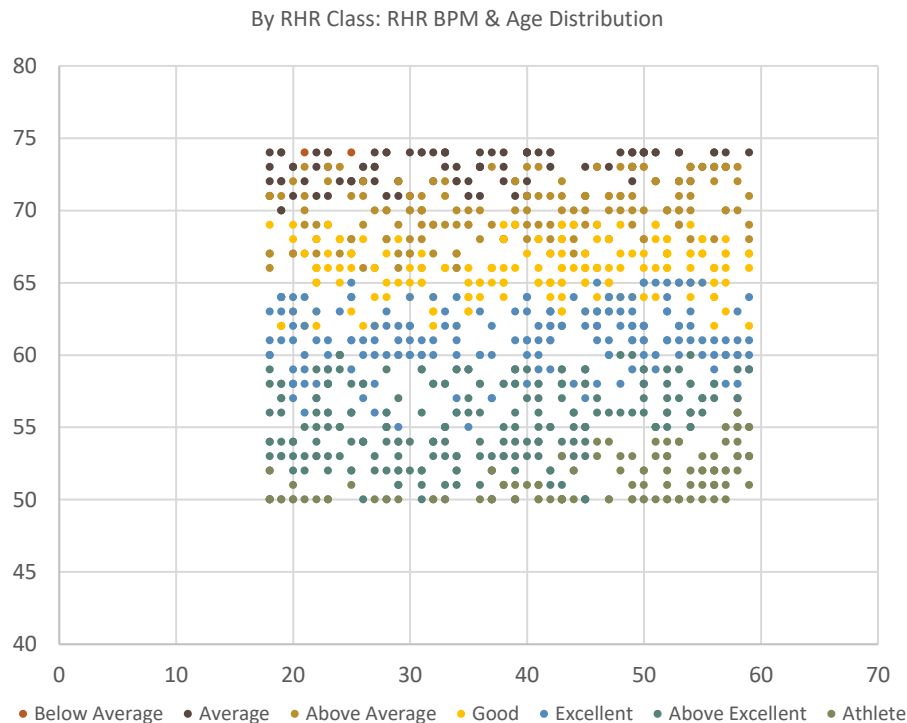
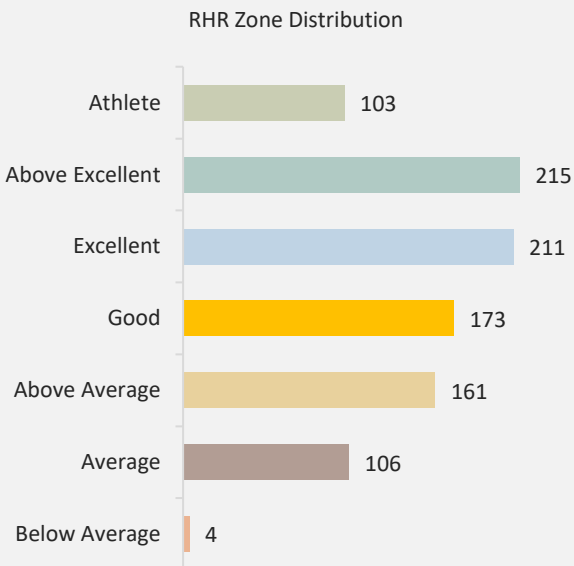
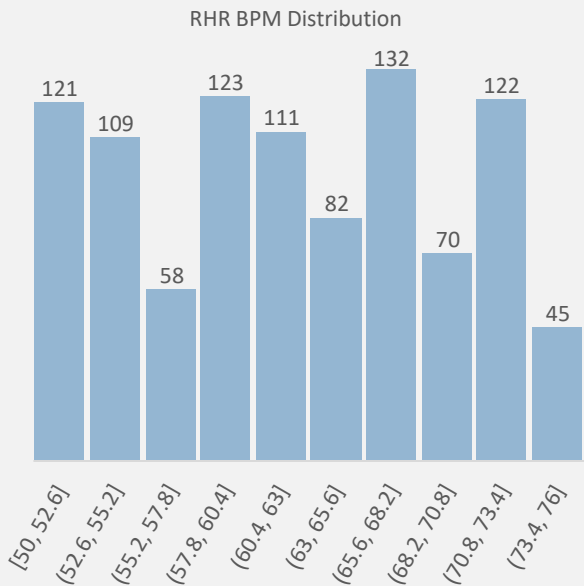
	Coefficients	P-value
Intercept	142.1281	0
Age	0.04236	0.262329

Insight

- The distribution of Avg EHR BPM is evenly distributed with no skew.
- 98.66% gym members reach normal and high EHR zones during their workout.
- For balance of 11.44% or 13 members, we found that they are inside category of critical EHR zone.
- Gym members who inside critical zone is within age of 55 – 60 years old with diversity of BMI class from severely underweight to obesity class 3.
- Regression analysis being done and found that there are weak statistic relationship between age and Avg EHR BPM with P-Value of 0.262 and the variance of Avg EHR BPM that can be explained by age of member is too low at 0.13%.
- Conclusion, to provide attention on these group of 13 members to ensure they are not pushing themselves to dangerous limit during workout to avoid severe health risk.
- EHR zone is impacted by age of person. The older they are, the lower their max zone level. (Source: <https://www.heart.org/en/healthy-living/fitness/fitness-basics/target-heart-rates>).

Proposal

- Conduct health assessment periodically for gym members to track their overall performance and fitness metric.
- Educate gym member on importance of monitoring heart level; how to ensure they are not jeopardizing their health during workout.



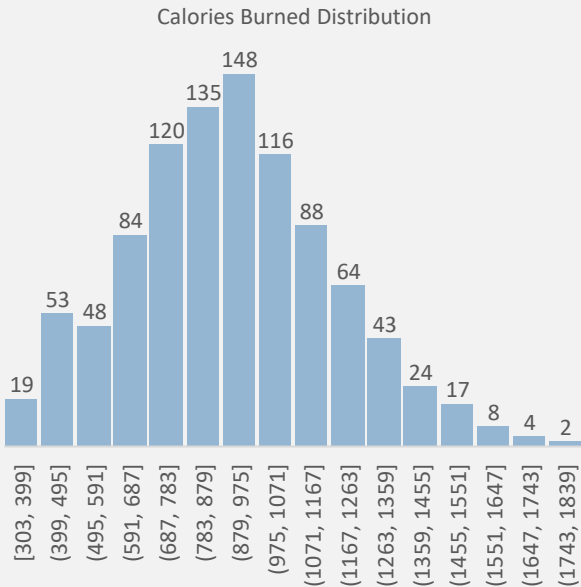
Independent Variable	Coefficients	P-value
Intercept	62.55886	0
Workout Frequency	-0.25306	0.591126
Experience Level	0.278882	0.631489

Insight

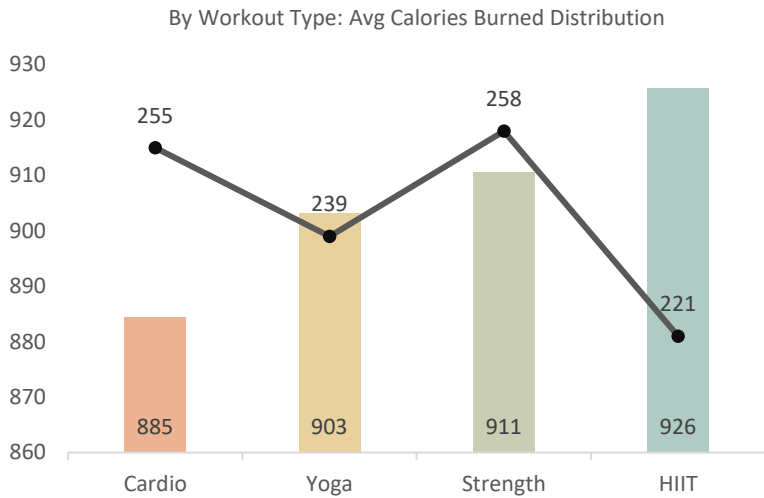
- The distribution of RHR BPM is evenly distributed.
- By RHR Zone, 72.15% of gym members are categorized within good and athlete zones.
- Of the 103 members inside Athlete category, 75 of them are males and balance of 28 are females.
- By fat class, we can see the diversity of the class for each RHR zone. Focus on Athlete zone, 29% of them are inside Very High fat class. This might due to various factors such as sports they are into, genetics and so on. Extensive research need to be done to find the accurate answer for each members. (Source: [Elite Body Fat Percentages | BodyStack](#)).
- By age distribution, we can see that each age group included inside each RHR zone. Indicating that age is not a major factor that influence RHR BPM.
- By correlation values, we found that none of the metrics in the dataset correlate with RHR BPM. A regression analysis also has been done with workout frequency and experience level as independent variable but the result shows that the significance statistic are 0.59 and 0.63 respectively which is a weak relationship.
- Indeed, need to consider other factors that might affect RHR BPM such as, comprehensive fitness level, body temperature, emotions, medications, substance use and etc. (Source: [Resting heart rate – definition, impacting factors & how to check it | healthdirect](#)).
- Bear in mind that RHR zone differ by age group and gender. (Source: https://www.medicinenet.com/what_is_a_good_resting_heart_rate_by_age/article.htm).
- Lower RHR BPM doesn't mean that they are in absolute best of fitness level as health problem also can cause RHR in lower zone. (Source: [Your resting heart rate can reflect your current and future health - Harvard Health](#)).

Proposal

- Conduct health assessment periodically for gym members to track their overall performance and fitness metric.
- Educate gym member on importance of monitoring heart level; how to ensure they are not jeopardizing their health during workout.

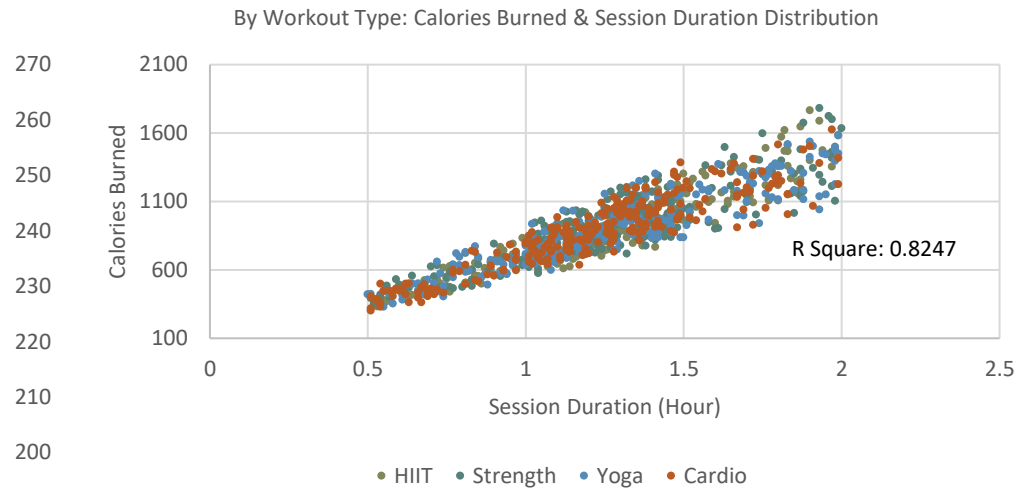


Correlation Matrices		Calories Burned
Calories Burned		1
Experience Level		0.694129448
Workout Frequency (Day / Week)		0.576150125
Session Duration (Hours)		0.908140376
Max BPM		0.002090016
Avg BPM		0.339658667
Water Intake (Liters)		0.356930683



Regression Statistics	
Multiple R	0.970504849
R Square	0.941879662
Adjusted R Square	0.941518666
Standard Error	65.93266412
Observations	973

Independent Variable	Coefficients	P-value
Intercept	-994.9282167	5.37E-98
Experience Level	-1.846706019	0.767857
Workout Frequency (Day / Week)	-3.287144295	0.438699
Session Duration (Hours)	701.0817453	0.0000
Max BPM	0.054842053	0.765744
Avg BPM	6.192662704	8.2E-220
Water Intake (Liters)	50.8609612	4.04E-39

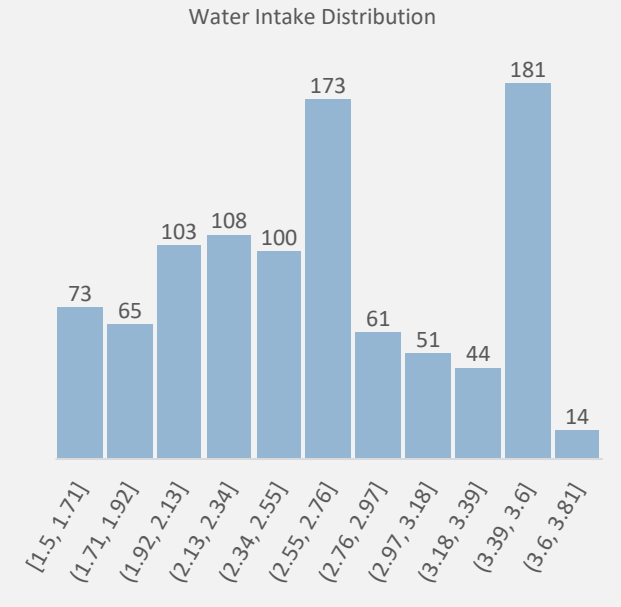


Insight

- Distribution of calories burned for all gym members is slightly skewed to the right.
- Correlation matrices for calories burned shows that all the metrics have value above 0.3 except for Max BPM at 0.002.
- High Intensity Interval Training or HIIT emerge as the leaders of average calories burned per member despite having the lowest number of members that participate, count at 221.
- The distribution varies evenly around hours 0.5 to 1.49, but the pattern diverse around 1.5 to 2 hours with some of them exceed 1600 calories burned.
- We can conclude that each workout will burned more calories when they exceed 1 hour and 30 minutes. This is because of heart rate will keep increasing the longer we did an exercise and the continual pumping of blood by the heart demands energy, as does the continual ventilation (movement of air into and out) of the lungs. (Source: [Calorie Burning](#)).
- With the result of multiple variable of regression analysis which we combine metrics of fitness, behavioural and performance, the result shows that session duration (hour) has the most impact by coefficient of 701 units increased per hour. Follow by water intake at 50.86 calories per litre of water intake and 6.19 calories increased per 1 avg EHR BPM.
- Experience level, workout frequency and max BPM has a weak relationship comparing to other variables which is the result not tally with their correlation values at 0.69 for experience level and 0.58 for workout frequency.
- Their impact is less significant in the regression model due to the presence of more influential variables and potential multicollinearity.

Proposal

- Enhanced training programs by emphasizing longer workout (1.5 hours and above) to maximise calories burning.
- To promote program of HIIT to other gym members with data showcasing that even with less participant, it is more effective to burn calories. But, this doesn't mean we can reject the effectiveness of other workouts.



Independent Variable	Coefficients	P-value
Intercept	5.751882	1.5E-48
Fat Percentage	-0.055	2.96E-52
Experience Level	-0.0597	0.196119
Workout Frequency (Day / Week)	-0.01058	0.723108
Session Duration (Hours)	-1.38028	6.18E-16
Max BPM	0.001082	0.403893
Avg BPM	-0.01238	1.02E-12
Resting BPM	-0.00106	0.604358
Calories Burned	0.001963	3.15E-18

